

RESEARCH IDEA DISCUSSION

Barrier-Based Platoon Control for Autonomous Intersection Management

Crossing-order decisions and multi-conflict safety filtering

Open-road intersections / V2X or roadside coordination

Proposal stage: method and experiments to be developed

The open-road coordination problem

Each new arrival can change who should cross first

Traffic interactions

- Several streams approach the intersection.
- One vehicle can encounter more than one conflict region.
- Following vehicles also need safe longitudinal gaps.

Control objective

- Coordinate crossing before vehicles enter the conflict regions.
- Preserve smooth motion when feasible.
- Evaluate delay and comfort under explicit safety constraints.

Context: reservation-based AIM is established in [1] [Dresner & Stone, 2008](#). Performance benefits here remain hypotheses.

Starting point and research gap

Evidence status	What it contributes
Verified literature	Reservation-based intersection coordination [1]; CBF constraints in optimization-based control [2].
Current draft	Describes a figure-eight barrier controller and a BaIP CARLA evaluation. Original sources still need verification.
Proposed extension	Open arrivals, directed movement conflicts, consistent crossing order and bounded acceleration.

Can order-aware filtering handle multiple conflicts without excessive stopping?

[1] Dresner & Stone, 2008; [2] Ames et al., 2017. No numerical benefit or novelty claim is inferred from the unverified reports.

Research questions and potential contributions

1. Consistent decisions

Can one crossing order resolve interacting conflicts without repeated yielding?

2. Feasible safety constraints

Under what initial states and actuator limits can all constraints be satisfied?

3. Traffic performance

What delay and comfort costs accompany stricter safety enforcement?

Potential contribution: a carefully defined order/filter interface,
feasibility analysis and reproducible evaluation.

A conflict graph or a CBF-QP alone is not claimed as a new contribution. Novelty requires a focused literature review.

Case A: three directed movements

Bidirectional road

$$\ell_1 = N \rightarrow S, \quad \ell_2 = S \rightarrow N$$

Unidirectional road

$$\ell_3 = E \rightarrow W$$

Fixed straight paths; separate opposing lanes.

Finite vehicle footprints define conflict regions, with path-specific entry and exit positions.

Proposed geometry, not experimental data. First study: all vehicles controlled, known routes, synchronized shared state.

Movement conflict matrix

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$A_{pq} = 1$: movements share a conflict region.

Two local conflicts for ℓ_3 ; no lateral conflict between ℓ_1 and ℓ_2 . Same-lane rear-end safety is handled separately.

Proposed control architecture

Stage	Information or output
Shared state	Path position, speed, route, dimensions and timestamp.
Conflict activation	Approaching and occupying vehicles; relevant movement pairs.
Decision layer	One consistent precedence relation across conflict regions.
Nominal control	Desired speed and same-lane headway acceleration u_i^{nom} .
Joint CBF-QP	Acceleration vector $\mathbf{u}$ satisfying active constraints, if feasible.
Execution	Apply inputs; monitor occupancy, feasibility and actual response.

First implementation: one roadside coordinator.

Proposed architecture. Distributed control is deferred because pairwise constraints couple both vehicles' inputs.

Decision layer: a stable crossing order

Rank eligible vehicles, then keep committed decisions

- Moving vehicles: rank by estimated arrival at the first relevant region. Near ties: approach-entry time, then vehicle ID.
- Preserve lane FIFO. Promote long-waiting vehicles only within the uncommitted set.
- Derive all conflict-pair precedences from one global order to avoid cyclic pairwise priorities.
- Freeze committed precedences until rear clearance. Admit a new order only after checking safety margins and feasibility.

Acyclic priorities do not by themselves prove deadlock freedom.

Low-speed admission and infeasible orders need explicit handling.

Barrier layer: filter the nominal acceleration

For moving vehicles approaching region m , with i before j :

$$T_i^m = \frac{d_i^m}{v_i}, \quad h_{ij}^m = T_j^m - T_i^m - \tau_m$$

$$\begin{aligned} \mathbf{u}^* = \arg \min_{\mathbf{u}} \quad & \sum_i w_i (u_i - u_i^{\text{nom}})^2 \\ \text{s.t.} \quad & \dot{h}_{ij}^m(\mathbf{u}) + \alpha h_{ij}^m \geq 0 \quad \text{for all active conflicting pairs,} \\ & u_{\min} \leq u_i \leq u_{\max}, \quad \text{plus speed and rear-end limits.} \end{aligned}$$

d_i^m : path distance to a conflict reference point; τ_m : proposed time gap.

Candidate formulation: arrival-time separation alone does not certify that vehicle occupancy intervals are disjoint.

CBF-QP foundation: [2] Ames et al., 2017. Requires $v_i, v_j > v_{\text{switch}} > 0$, fixed order and a feasible safe initial state.

Conditions that determine feasibility

Issue	Required treatment
Low or zero speed	d/v is unsuitable; use a stopping/admission mode with checked transitions.
Finite occupancy	Track front entry and rear exit; retain conflicts until clearance.
Bounded braking	Define a viable approach envelope and check simultaneous constraints.
QP infeasibility	Log the event; trigger a validated backup when available. Do not silently soften safety constraints.
Order and time steps	Check new safe-set membership at switches and motion between samples.

Safety is a conditional research target, not an established result.

Experimental design

Comparisons

- Actuated signal control.
- Unsignalized priority control.
- Original distance barrier, only in its supported setting.
- Fixed order vs. proposed order; heuristic vs. QP filtering.

Measurements

- Collisions, occupancy overlap, minimum separation.
- Throughput, delay, unfinished trips.
- Stops, acceleration and jerk RMS.
- Deadlock, infeasibility and solve time.

Matched arrivals and seeds across low, medium and high demand; include startup transients and asymmetric flows. Energy is conditional on a specified model.

Platform: [\[3\] Eclipse SUMO documentation](#). Audit simulator safety interventions so that they do not conceal controller failures.

Milestones and completion criteria

Milestone	Evidence required to advance
1. Model / baselines	Source audit, geometry, information assumptions and reproducible baseline runs.
2. Minimal closed loop	Case A stress cases; checked input bounds, order transitions, occupancy and failure logs.
3. Comparative study	Paired-seed comparisons, ablations, uncertainty estimates and reported adverse cases.
4. Extension and limits	Bidirectional crossroad, then turns; delay/error sensitivity. Optional CARLA validation.

Milestone-based progression; multi-intersection networks remain future work.

Questions for discussion

1. Contribution

Is the order/filter interface and feasibility analysis a strong enough core?

2. Theory target

Should the first result focus on a restricted safe operating domain or a richer occupancy-based formulation?

3. Scope

After Case A, should we prioritize turning movements or communication uncertainty?

Backup 1: dynamics and the coupled constraint

Let $\dot{d}_i^m = -v_i$, $\dot{v}_i = u_i$. For positive speeds, fixed order and constant τ_m :

$$\dot{T}_i^m = -1 - \frac{d_i^m}{v_i^2} u_i, \quad \dot{h}_{ij}^m = \frac{d_i^m}{v_i^2} u_i - \frac{d_j^m}{v_j^2} u_j.$$

$$\frac{d_i^m}{v_i^2} u_i - \frac{d_j^m}{v_j^2} u_j + \alpha \left(\frac{d_j^m}{v_j} - \frac{d_i^m}{v_i} - \tau_m \right) \geq 0.$$

Same-lane candidate: $h_f = g_f - s_0 - t_h v_f$, hence

$$\dot{h}_f = v_{\text{lead}} - v_f - t_h u_f.$$

Continuous-time invariance needs valid CBF conditions, $h(0) \geq 0$ and feasible inputs throughout. New orders, low-speed modes and sampled execution need separate arguments.

Independent algebra for this proposal; general invariance framework: [\[2\] Ames et al., 2017](#). g_f is bumper gap; $t_h > 0$.

Backup 2: ordered and unordered margins

Unordered margin

$$h = |T_i - T_j| - \tau$$

Allows either order. The two safe branches are separated; the absolute value is nonsmooth at equal arrival times.

Ordered margin

$$h = T_j - T_i - \tau$$

Encodes i before j . Switching order changes the safe set and requires an explicit admissibility check.

Two heuristic pitfalls in the draft

$\phi(h) = -1/h$ is singular at zero and changes sign for $h < 0$.

For negative braking corrections, $\max(-3, -1) = -1$ selects weaker braking. Neither a sum nor an extremum automatically satisfies all safety constraints.

Backup 3: evaluation definitions

Measure	Definition and reporting rule
Safety	Geometric contact/overlap; region occupancy overlap; minimum ordered margin. Report separately.
Efficiency	Exits per hour; delay against same-route free flow, including departure delay; unfinished/uninserted vehicles.
Comfort	Acceleration RMS (m/s^2), jerk RMS (m/s^3), stops per vehicle; report sampling interval.
Liveness	Max waiting time; blocked demand with no crossing for 60 s is a deadlock flag for inspection.
Feasibility	QP failure rate, backup activations, input violations; median and 95th-percentile solve time.
Optional measures	Trajectory-based collision-time estimate, distinct from d/v ; energy only with stated model and units.

Proposed evaluation definitions. Simulator output interpretation: [SUMO TripInfo](#); [SUMO Safety](#).

Backup 4: verified sources and open provenance

[1] K. Dresner and P. Stone (2008)

A Multiagent Approach to Autonomous Intersection Management.

JAIR 31, 591–656. [Author-hosted paper](#).

[2] A. D. Ames, X. Xu, J. W. Grizzle and P. Tabuada (2017)

Control Barrier Function Based Quadratic Programs for Safety Critical Systems. IEEE TAC 62(8), 3861–3876.

[Caltech record](#); DOI: 10.1109/TAC.2016.2638961.

[3] Eclipse SUMO documentation

[TraCI vehicle control](#), [safety](#), [traffic lights](#), [trip outputs](#). Accessed 24 September 2026.

Open provenance: figure-eight theory and BalP report mentioned in the local draft. Original publications and quantitative claims remain unverified.